

Supply Chain Operations & Performance Analysis Report

Executive Summary

Efficient supply chain management requires complete visibility into inventory, suppliers, warehouse operations, customer demand, and business profitability. This project analyzes operational supply chain data to identify key business challenges and provide data-driven recommendations that improve operational efficiency, reduce costs, and support strategic decision-making.

The analysis focuses on five major business areas that directly impact supply chain performance and overall business growth.

Business Problem 1

Inventory Shortage & Overstock Analysis

Problem

The company lacks visibility into inventory levels across warehouses. Some products frequently fall below the reorder point, causing stock shortages, while others remain overstocked, increasing inventory carrying costs and reducing warehouse efficiency.

Solution

Inventory levels were analyzed across warehouses and regions to identify low-stock products, overstocked inventory, and warehouses requiring immediate replenishment.

Output

- Identified warehouses containing the highest number of low-stock products.
- Identified warehouses with excessive inventory.
- Compared inventory shortages across different regions.
- Detected the top overstocked products requiring inventory optimization.

Business Impact

- Reduced stockout risk.
- Lower inventory holding costs.
- Better warehouse space utilization.
- Improved inventory planning.

Recommendation

Implement automated inventory monitoring and reorder alerts while redistributing excess inventory between warehouses to maintain balanced stock levels.

Business Problem 2

Supplier Performance Analysis

Problem

The procurement department lacks sufficient visibility into supplier performance. Delivery delays and supplier dependency can negatively affect inventory availability and overall supply chain efficiency.

Solution

Supplier performance was evaluated using delivery lead time, total order quantity, and product coverage to identify both high-performing and underperforming suppliers.

Output

- Supplier lead times ranged between approximately 7 and 9 days.
- **SUP_4** recorded the highest average lead time, making it the slowest supplier.
- **SUP_5** demonstrated the fastest delivery performance.
- **SUP_7** supplied the highest order quantity and the widest variety of products.

Business Impact

- Faster supplier selection.
- Reduced procurement delays.
- Improved inventory availability.
- Lower supplier dependency risk.

Recommendation

Prioritize high-performing suppliers for critical inventory, continuously monitor supplier KPIs, and diversify procurement sources to reduce operational risk.

Business Problem 3

Warehouse Performance Analysis

Problem

Management requires better visibility into warehouse operations to optimize inventory allocation, warehouse capacity, and operational efficiency.

Solution

Warehouse performance was evaluated by comparing sales volume, inventory levels, order quantities, and regional sales performance.

Output

- Warehouse sales remained relatively balanced across all locations.
- **WH_1** achieved the highest sales volume.
- **WH_2** maintained the highest inventory and processed the largest order quantity.
- The **East** region generated the highest total sales.

Business Impact

- Improved warehouse performance monitoring.
- Better inventory allocation.
- Increased operational efficiency.
- More effective warehouse resource planning.

Recommendation

Adopt operational best practices from high-performing warehouses and continuously monitor warehouse KPIs to maintain balanced inventory and workload distribution.

Business Problem 4

Demand Forecasting Analysis

Problem

Accurately forecasting customer demand is essential for preventing stock shortages, reducing excess inventory, and improving purchasing decisions.

Solution

Demand forecasts were compared with actual sales while evaluating regional demand patterns, product demand, and the impact of promotional campaigns.

Output

- Forecast demand closely matched actual sales across most products.
- The **East** region showed the highest forecasted demand.
- Promotional campaigns significantly increased average units sold.
- Several products consistently exhibited high forecast demand.

Business Impact

- Improved demand planning.
- Better purchasing decisions.
- Reduced inventory waste.
- Increased product availability.

Recommendation

Continuously improve demand forecasting models, maintain sufficient inventory for high-demand products, and align promotional campaigns with forecasted demand patterns.

Business Problem 5

Revenue & Profitability Analysis

Problem

The company needs to identify which products, warehouses, and regions generate the highest financial returns in order to optimize pricing strategies, inventory investment, and business growth.

Solution

Revenue and profitability were evaluated across warehouses, regions, and individual products to determine the primary contributors to business performance.

Output

- **WH_2** generated the highest overall revenue and profit.
- **SKU_38** emerged as the most profitable product.
- The **North** region generated the highest revenue.
- High-profit products contributed significantly to overall business profitability.

Business Impact

- Improved financial visibility.
- Better pricing decisions.
- Optimized inventory investment.
- Higher business profitability.

Recommendation

Focus inventory investment and marketing efforts on high-profit products, regularly monitor profitability metrics, and expand successful sales strategies across all business regions.

Overall Business Outcome

This analysis provides a comprehensive view of supply chain operations by identifying inventory risks, evaluating supplier performance, measuring warehouse efficiency, assessing customer demand, and analyzing financial performance.

The insights generated through this project enable organizations to improve operational efficiency, reduce unnecessary inventory costs, strengthen supplier management, optimize warehouse operations, enhance demand planning, and maximize overall business profitability through data-driven decision-making.

Conclusion

The **Supply Chain Operations & Performance Analysis Report** demonstrates how operational data can be transformed into actionable business intelligence. By addressing five critical business challenges with analytical solutions and practical recommendations, the project supports informed decision-making across inventory management, procurement, warehouse operations, demand planning, and financial performance.

This project reflects a real-world supply chain analytics workflow and showcases practical business analysis techniques commonly used by organizations to improve operational performance and drive sustainable growth.